

AMIT MAHENDRABHAI VAGHELA

AI Data Analyst | Data Analyst

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PROFESSIONAL SUMMARY

- Results-driven AI Data Analyst with 3+ years of experience transforming complex datasets into actionable business insights through advanced analytics, machine learning, data visualisation, and automation solutions.
- Strong expertise in Python, SQL, Pandas, NumPy, SciPy, Scikit-learn, TensorFlow, and XGBoost, leveraging statistical analysis, predictive modelling, classification, regression, feature engineering, and model evaluation techniques to solve business problems.
- Experienced in building AI and machine learning solutions, including time series forecasting, ensemble learning, hyperparameter tuning, explainable AI (SHAP), cross-validation, and optimisation techniques to uncover patterns and support data-driven decision-making.
- Skilled in developing interactive dashboards and business intelligence solutions using Power BI, Excel, Plotly, and Streamlit, creating KPI reporting frameworks, data visualisations, exploratory data analysis (EDA), and automated reporting solutions for stakeholders.
- Hands-on experience with data engineering concepts, including ETL pipelines, structured data processing, SQL analytics, data cleaning, data modelling, dimensional modelling, star schema design, and data quality validation using SQL Server, PostgreSQL, MySQL, DuckDB, and Parquet-based data processing.
- Experienced in designing scalable analytical workflows by integrating data pipelines, automation processes, and AI solutions while maintaining data accuracy, audit logging, documentation, and governance standards.
- Familiar with modern AI/LLM technologies, including Retrieval-Augmented Generation (RAG), Graph RAG, Hybrid RAG, Agentic AI workflows, Multi-Agent Systems, Prompt Engineering, LangChain, LangGraph, Hugging Face, Pinecone, and Sentence Transformers for building intelligent data-driven applications.
- Proficient in developing lightweight analytical applications using FastAPI, Streamlit, React, Next.js, TypeScript, JavaScript, and Plotly Dash, enabling interactive user experiences and self-service analytics capabilities.
- Strong understanding of business analysis practices, including requirements gathering, gap analysis, process mapping, workflow documentation, Agile methodologies, JIRA, Confluence, and stakeholder communication to translate business needs into technical solutions.
- A solution-oriented analytics professional with experience collaborating with cross-functional teams to automate processes, improve operational efficiency, identify trends, and deliver measurable business value through data and AI-driven insights.

WORK EXPERIENCE

AI Data Analyst | Microsoft, Canada | Jan 2025 – till date

- Analysed complex business datasets using Python, SQL, Pandas, NumPy, and statistical techniques to identify trends, generate actionable insights, and support data-driven decision-making across business teams.
- Designed and developed interactive Power BI dashboards and reports using DAX measures, calculated columns, KPI frameworks, and data visualisation techniques to monitor business performance and operational metrics.

- Built and optimised ETL workflows using SQL, Power Query, and Azure data services to extract, transform, validate, and integrate data from multiple enterprise sources for analytics and reporting.
- Developed scalable data models using Star Schema and dimensional modelling approaches, improving reporting accuracy, dashboard performance, and enabling self-service analytics for stakeholders.
- Applied machine learning techniques, including predictive analytics, classification, regression, and feature engineering using Python and Scikit-learn to identify patterns, forecast trends, and support business optimisation.
- Automated reporting and data processing workflows using Power Automate, Python scripting, and Power BI scheduled refresh, reducing manual effort and improving reporting efficiency.
- Implemented data quality checks, validation processes, and governance practices to maintain accurate, reliable, and secure datasets across analytical solutions.
- Collaborated with business stakeholders, product teams, and engineering teams to gather requirements, define KPIs, translate business challenges into analytical solutions, and deliver AI-driven insights.

Data Analyst | BG fuels |Jan 2024 - Oct 2024

- Analysed fuel sales, operational, and transactional datasets using SQL, Excel, and Power BI to identify trends, monitor business performance, and support data-driven decision-making.
- Developed and maintained interactive Power BI dashboards and reports to track key business metrics, including sales performance, fuel consumption trends, revenue analysis, operational efficiency, and KPI performance.
- Extracted, cleaned, and transformed data from multiple sources, including databases, Excel files, and operational systems, using SQL and Power Query to improve data accuracy and reporting reliability.
- Built data models and created DAX measures, calculated columns, and KPI calculations to provide actionable insights into sales trends, customer behaviour, and operational performance.
- Performed data validation, reconciliation, and quality checks to identify inconsistencies, resolve data issues, and ensure accurate reporting across business functions.
- Conducted trend analysis and variance analysis on fuel sales, pricing patterns, and operational metrics to identify improvement opportunities and support business planning.
- Collaborated with operations, finance, and business stakeholders to gather reporting requirements, define KPIs, and deliver analytical solutions that improved visibility into business performance.

Data Analyst | BG Fuels| Nov 2021 – Jan 2024

- Conducted sales, revenue, and operational performance analysis by analysing large transactional datasets to identify business trends, cost drivers, and opportunities for process improvement.
- Performed forecasting and trend analysis on fuel consumption, inventory movement, and customer demand patterns to support planning activities and improve operational efficiency.
- Delivered ad-hoc analytical reports and business insights by combining data analysis, statistical techniques, and advanced Excel capabilities to support management decisions and strategic initiatives.
- Worked with large transactional datasets to perform data profiling, anomaly detection, and reconciliation, ensuring accuracy before presenting analytical findings.
- Documented analytical processes, reporting procedures, and business rules to improve knowledge sharing and maintain consistency across reporting activities.

TECHNICAL SKILLS

- **Programming & Analytics:** Python, SQL, R, Pandas, NumPy, SciPy, Excel, VBA, Statistical Analysis, EDA, Data Cleaning, Data Transformation
- **AI/ML:** Machine Learning, Predictive Analytics, NLP, Time Series Forecasting, Classification, Regression, Feature Engineering, XGBoost, LightGBM, CatBoost, TensorFlow, PyTorch, Explainable AI
- **Generative AI:** LLMs, Generative AI, RAG, Agentic AI, Prompt Engineering, LangChain, LangGraph, Hugging Face, Pinecone, Ollama
- **BI & Visualisation:** Power BI, Tableau, DAX, Power Query, KPI Reporting, Dashboards, Data Visualisation, Reporting Automation
- **Data Engineering:** ETL/ELT, Data Pipelines, Data Modelling, Star Schema, Dimensional Modelling, Data Quality, SQL Analytics, Data Warehousing
- **Cloud & Tools:** Azure, AWS, Snowflake, SQL Server, PostgreSQL, MySQL, Docker, GitHub, Jupyter Notebook, Streamlit, FastAPI
- **Business Skills:** Requirements Gathering, Stakeholder Management, Agile, JIRA, Confluence, Process Mapping, Workflow Documentation

EDUCATION

PG in Diploma in Logistics & Supply Chain Management, Canadore College, Canada	2020-2021
PG in Construction, Canadore College, Canada	2019-2020
Bachelor's in Civil Engineering, Gujarat Technological University, India	2014-2018

CERTIFICATIONS

- Google Advanced Data Analytics, Google.
- Google Data Analytics, Google.
- Google AI Fundamentals, Google.
- AI for Data Analysis, Google.
- AI for Research and Insights, Google.
- AI for Brainstorming and Planning, Google.
- AI for Writing and Communicating, Google.
- The Nuts and Bolts of Machine Learning, Google.